MODULE 5.1 · Concept 7 of 8 · NEW

SLMs as Lightweight Agent Engines

The industry trend: big brain for planning, small brain for execution — 3–10× cost savings

8 min	B3 – Apply	L2 – Intermediate	TH-000006
Duration	Bloom's	Level	Prerequisite

AI-AG-FN-TH-000019

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SECTION 1 — WHY

The Cost-Latency Problem

Using GPT-4o for everything is like hiring a surgeon to take your temperature

The Problem: Uniform Model Selection Wastes Money

Not every agent step requires the most powerful (and expensive) model

An 8-step agent task using GPT-4o for every step:

Step 1: Parse user intent	**Step 2:** Search flights API	**Step 3:** Format API response	**Step 4:** Compare options
Step 5: Select best option	**Step 6:** Generate booking call	**Step 7:** Confirm with user	**Step 8:** Log resolution

- Only steps 1 and 5 (green) require complex reasoning. The other 6 are simple formatting, routing, and logging.
- Yet all 8 pay \$0.03 each = \$0.24/task.

At 100K tasks/day: \$0.24 × 100K = \$24,000/day = \$8.76M/year. Most of it wasted on simple steps.

Tiered Model Routing — The Pattern

Big models plan, small models execute

Tier 1: Planning GPT-4o / Claude Opus	Complex reasoning, task decomposition, ambiguity resolution. Only 2 of 8 steps need this.	\$0.03/call ~800 ms
Tier 2: Tool Calls GPT-4o-mini / Haiku	JSON formatting, API parameter extraction, simple decisions. 4 of 8 steps.	\$0.001/call ~200 ms
Tier 3: Classification Phi-3 / Llama 3 8B	Intent routing, binary yes/no decisions, guard checks. 2 of 8 steps.	\$0.0001/call ~50 ms

The key insight: match model capability to step complexity. Don't use a \$0.03 model for a \$0.0001 task.

Tier 1 Deep Dive: Planning with Large Models

When you DO need the most powerful model

Tier 1 tasks require:

- Multi-step reasoning across ambiguous constraints
- Task decomposition (breaking 'plan my vacation' into 15 sub-tasks)
- Resolving conflicting goals ('cheapest' vs 'most comfortable')
- Understanding nuanced user intent ('make it more professional but keep the jokes')
- Planning recovery from failures ('Plan A failed, generate Plan B')

Models for Tier 1:

GPT-4o — Best general reasoning

Claude Opus 4 — Best code + safety

Gemini Ultra — Longest context (2M)

Llama 3 70B — Best open-source option

Rule of thumb: if the step requires >3 seconds of human thought, use Tier 1.

Tier 2-3 Deep Dive: Small Models for Execution

Most agent steps don't need complex reasoning

Tier 2: Tool Call Formatting

What: Convert a planned action into structured JSON
Why small is enough: The PLAN already exists from Tier 1.
The model just needs to format parameters.

Examples:
- Extract API parameters from planned text
- Format a database query from specifications
- Generate a structured email from a draft

Models: GPT-4o-mini, Claude Haiku, Mistral 7B

Tier 3: Classification & Guards

What: Binary or multi-class decisions
Why small is enough: Choosing from fixed categories
doesn't need reasoning — just pattern matching.

Examples:
- Intent classification (5 categories)
- Toxicity detection (safe/unsafe)
- Language identification

Models: Phi-3 3.8B, TinyLlama 1.1B, Gemma 2B

Cost Calculation: 3.7× Savings with Tiered Routing

Worked example — 8-step agent task

Uniform Approach (All GPT-4o):

8 steps × \$0.03/step = \$0.24 per task

At 100K tasks/day: \$24,000/day = \$8.76M/year

Tiered Approach:

```
2 steps @ Tier 1 (GPT-4o):      2 × $0.030 = $0.060
4 steps @ Tier 2 (GPT-4o-mini):  4 × $0.001 = $0.004
2 steps @ Tier 3 (Phi-3):        2 × $0.0001 = $0.0002
```

Total: \$0.0642 per task

Savings: \$0.24 → \$0.064 = 3.7× cheaper

At 100K tasks/day: \$24,000 → \$6,420/day

Annual savings: \$6.4M

SECTION 2 — THE SLM LANDSCAPE

Small Language Models for Agents

Choosing the right model for each agentic subtask

SLM Landscape: Five Models Compared

Parameters, VRAM, latency, and best use case for each

Model	Params	VRAM	p50 Latency	Best Agentic Use
Phi-3 Mini	3.8B	~8 GB	~50 ms	On-device classification, intent routing
Llama 3 8B	8B	~16 GB	~80 ms	Tool call formatting, JSON extraction
Mistral 7B	7B	~14 GB	~70 ms	Code generation, structured output
TinyLlama	1.1B	~2 GB	~15 ms	Guard agents, simple binary decisions
Gemma 2B	2B	~4 GB	~25 ms	Mobile/edge agents, text classification

All models can run on a single GPU. TinyLlama and Gemma can run on a smartphone.

Latency Calculation: 2.6× Faster User Experience

Speed matters as much as cost

6.4 sec	2.5 sec	2.6×
All GPT-4o (8 × 800ms)	Tiered routing (2×800 + 4×200 + 2×50)	Faster user experience

- For interactive agents, latency directly impacts user satisfaction. A 2.5-second response feels instant.
- A 6.4-second response feels slow. Tiered routing improves both cost AND experience.

The Cost-Accuracy-Latency Triangle

You can optimize two — the third is the tradeoff

COST

GPT-4o: High accuracy + low latency
→ High cost ($0.03)

TinyLlama: Low cost + low latency
→ Lower accuracy

ACCURACY

LATENCY

Tiered Routing: Balances all three
→ Best overall tradeoff

On-Device and Edge Agents

TinyLlama and Gemma enable agents without cloud dependency

Why on-device matters:

- Privacy: sensitive data never leaves the device (medical, financial, legal)
- Offline: agents work without internet (field workers, aircraft, remote areas)
- Latency: no network round-trip — response in <50ms
- Cost: zero API charges after model deployment
- Compliance: data residency requirements (GDPR, HIPAA)

Viable on-device models:

TinyLlama 1.1B (~2 GB RAM) — Runs on phone

Gemma 2B (~4 GB RAM) — Runs on tablet

Phi-3 Mini 3.8B (~8 GB RAM) — Runs on laptop

- Pattern: Use on-device SLM for Tier 3 tasks (classification, guard checks, intent routing).
- Escalate to cloud LLM only for Tier 1 tasks that require complex reasoning.

Designing a Tiered Agent Pipeline

Decision flowchart: which model for which step

Does this step require multi-step reasoning or ambiguity resolution?	**YES → Tier 1 (GPT-4o / Claude Opus)** NO ↓ Continue
Does this step require structured output or tool call formatting?	**YES → Tier 2 (GPT-4o-mini / Haiku)** NO ↓ Continue
Is this step a classification, guard check, or binary decision?	**YES → Tier 3 (Phi-3 / TinyLlama)** NO ↓ Reactive rule

This flowchart runs at design time (pipeline setup) not at runtime. Each step's tier is fixed in the agent architecture.

Key Takeaways

The essential points from this concept

Tiered routing: big models plan (Tier 1), small models execute (Tier 2–3). Match model to task.

Cost impact: 3.7× cheaper. At scale (100K tasks/day), saves $17,600/day = $6.4M/year.

Latency impact: 2.6× faster end-to-end agent response (6.4s → 2.5s).

SLM sweet spot: Phi-3 for routing, Llama 3 8B for tool calls, TinyLlama for guards.

On-device agents: TinyLlama/Gemma enable privacy-first, offline, zero-cost edge deployment.

Next: AI-AG-FN-CD-000001 — Building a Simple Agent (hands-on LangChain coding lab)

AI-AG-FN-TH-000019 Complete

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Concept 7 of 8

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